

USING RESOURCES – LIFE CYCLE ASSESSMENT

KEY VOCABULARY

Life cycle assessment: measures the environmental impact of a product.

Raw materials: the resources extracted to make the product.

Manufacture & use: energy/water used making and using the product.

Disposal: what happens at the end — landfill or recycling.

COMMON MISCONCEPTIONS

~~An LCA only looks at making a product.~~

✓ It covers the whole life from raw materials to disposal.

~~LCAs are completely objective.~~

✓ Some impacts are hard to measure, so judgements are made.

~~Recycling has no environmental cost.~~

✓ Collecting and reprocessing also use energy.

FOUR LCA STAGES

1. Extracting and processing raw materials.

2. Manufacturing and packaging.

3. Using and operating the product.

4. Disposal at end of life.

Consider energy, water, resources and waste at each stage.

1 DEFINE LCA

State what a **life cycle assessment** measures. [1 mark]

2 FOUR STAGES

List the **four stages** considered in an LCA. [2 marks]

3 COMPARE BAGS

Explain why a plastic bag may have a lower LCA impact than a paper bag in some areas. [2 marks]

4 LIMITATIONS

Explain why an LCA is not entirely objective. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

An LCA runs from ____ all the way through to ____.

raw materials

disposal

profit

advertising

6 APPLY

Suggest how a company could use an LCA to improve a product. [1 mark]